Wall time per step

Grey: the step where the run died. Blank: step not available or not reached.

100k (69,412 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	cluster_fast	umap	dge	tsne_default	tsne_fft	tsne_quick
bixverse mtx	10	0.7		0.3	1.3	0.6	2.1	0.8	0.8	2.6	0.7	26	7.6	4.6
bixverse h5ad	6.0	0.8		0.3	1.2	0.6	2.1	1.0	0.8	2.6	0.7	26	7.9	4.8
bixverse GPU*	5.8	0.5		0.3	0.5	0.6	1.2	0.8	0.4	0.6	0.6	24	2.3	3.2
bixverse MLX*	5.5	0.4		0.3	0.4	0.4	1.1	0.7	0.3	0.5	0.6	24	2.1	2.8
Scanpy mtx	19	0.7	2.0	1.4	1.4	0.6	27	2.0		27	56	96	32	
Scanpy h5ad	2.3	0.8	1.9	1.4	1.4	0.6	27	2.0		27	52	95	32	
Scanpy dask	6.1	1.1	0.0	1.7	4.0	0.5	24	1.9		26		102	31	
scrapper mtx	38	0.5	0.1	0.2	1.7	12	4.3	8.8	10	32	6.2			17
scrapper h5ad	6.5	0.5	0.1	0.2	1.7	12	4.6	8.7	11	33	6.3			18
Seurat std mtx	43	2.6	0.1	4.0	7.1	13	5.3	18		35	15	66	24	
Seurat std h5ad	10	2.7	0.1	4.0	6.3	13	5.4	18		35	16	71	24	
Seurat BPCells	15	1.2	0.5	4.0	73	14	5.5	18		35	196	71	24	

500k (346,259 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	cluster_fast	umap	dge	tsne_default	tsne_fft	tsne_quick
bixverse mtx	55	1.9		0.9	2.5	2.1	11	4.0	12	22	2.6	150	37	24
bixverse h5ad	27	1.9		1.1	2.4	2.2	11	4.1	12	22	2.6	149	38	25
bixverse GPU*	30	1.4		0.8	0.9	1.3	5.8	3.3	1.9	2.5	2.4	132	9.5	14
bixverse MLX*	23	1.3		0.8	1.3	0.6	5.2	3.1	1.6	2.4	2.3	129	13	13
Scanpy mtx	OOM													
Scanpy h5ad	12	3.8	3.5	3.2	5.7	2.4	31	14		155	602	796	109	
Scanpy dask	23	4.0	0.0	5.0	9.8	2.3	27	13		153		806	109	
scrapper mtx	OOM													
scrapper h5ad	162	2.2	0.4	0.9	10	87	26	89	117	150	36			96
Seurat std mtx	OOM													
Seurat std h5ad	OOM													
Seurat BPCells	70	4.2	2.5	14	350	92	33	111		198	953	598	110	

full (684,932 cells)

	load	qc	normalise	hvg	pca	harmony	knn_snn	cluster	cluster_fast	umap	dge	tsne_default	tsne_fft	tsne_quick
bixverse mtx	102	3.6		1.6	4.4	3.6	24	8.3	38	50	4.6	368	116	82
bixverse h5ad	74	4.4		1.7	4.2	3.7	24	8.3	39	50	4.7	371	114	81
bixverse GPU*	64	2.5		1.5	3.4	2.2	12	6.9	4.8	5.6	4.2	320	20	41
bixverse MLX*	47	2.3		1.4	2.3	1.1	11	6.9	3.8	5.2	4.1	309	19	40
Scanpy mtx	OOM													
Scanpy h5ad	OOM													
Scanpy dask	40	7.4	0.0	8.5	17	4.5	33	30		349		1,941	247	
scrapper mtx	OOM													
scrapper h5ad	OOM													
Seurat std mtx	OOM													
Seurat std h5ad	OOM													
Seurat BPCells	143	9.4	4.6	27	703	218	70	258		451	1,348	1,531	229	

wall time: 1 s · 10 s · 100 s · 1,000 s

bixverse CPU tSNE builds its kNN with IVF at 500k and full (kmknn at 100k).